

Pulkit Verma

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Research Interests

AI Safety, AI Planning, Analysis of Abstractions, and Action-Model Learning.

Education

Ph.D in Computer Science, Arizona State University

Advisor: Prof. Siddharth Srivastava [✉](#)

Tempe, AZ
Expected Fall 2023

Master of Technology in Computer Science & Engineering, IIT Guwahati

Advisor: Prof. Pradip K. Das [✉](#) | Thesis: Resource Usage Analysis for Speech Recognition Techniques. [✉](#)

Guwahati, India
May 2015

Bachelor of Engineering in Information Technology, SVITS, Indore

Advisor: Prof. Anand Rajawat | Capstone: Smart Analyzer - Predict and Cache Webpages based on Usage.

Indore, India
June 2011

Publications

Pulkit Verma, Shashank Rao Marpally, and Siddharth Srivastava. "Asking the Right Questions: Learning Interpretable Action Models Through Query Answering". In *Proc. of the AAAI Conference on Artificial Intelligence*, 2021. (To appear) [✉](#)

Pulkit Verma, and Pradip K. Das. "A Comparative Study of Resource Usage for Speaker Recognition Techniques". In *International Conference on Signal Processing and Communication*, 2016. [✉](#)

Pulkit Verma, and Pradip K. Das. "i-Vectors in Speech Processing Applications: A Survey". In *International Journal of Speech Technology*, 18(4), pp.529-546, 2015. [✉](#)

Mayank Gupta, **Pulkit Verma**, Tuhin Bhattacharya, and Pradip K. Das. *A Mobile Agents based Distributed Speech Recognition Engine for Controlling Multiple Robots. 2nd International Conference on Advances In Robotics*, 2015. [✉](#)

Pulkit Verma, Mayank Gupta, Tuhin Bhattacharya, and Pradip K. Das. "Improving Services using Mobile Agents-based IoT in a Smart City". In *2014 International Conference on Contemporary Computing and Informatics*, 2014. [✉](#)

Professional Experience

Arizona State University

Graduate Research Associate

Tempe, AZ
Aug. 2018 - (present)

- Investigating the minimal set of interfaces in an AI system that would enable a user to assess and understand the limits of its safe operability.
- Developed a framework which generates interrogation policies to derive the human interpretable model of an autonomous agent.
- Reduced the STRIPS-model comparison to a planning problem which enabled identifying if two models are not same.
- Modified FF-planner to take PDDL models, initial state, and plan as input; and output the final state.
- Co-designed the experiments showcasing possible use-cases of an NSF project on training lay people to use adaptive AI systems which are safe, robust and reliable.

NetApp Inc.

Storage Efficiency Developer

Bengaluru, India
Jul. 2015 - Jul. 2018

- *Workload Prediction*: Predicted workload on a storage volume based on I/O patterns for All Flash systems. This helped in optimizing data storage and management while keeping the nature of data private to the users.
- *File and Volume Clones*: Developed features and managed issues related to file and volume clones on UNIX-based NetApp proprietary WAFL file system to improve the storage efficiency of enterprise data servers.
- Co-developed a file system component which prevented data loss if files and volumes are cloned or modified when cloud backup is offline.

Tata Consultancy Services Ltd.

iOS Application Developer

Pune, India
Mar. 2012 - Aug. 2013

- Built interactive applications for iPhone and iPad and developed tools to analyze the user investment portfolio.
- Performed cross-platform application development of mobile applications supported on both Android and iOS.

Teaching Experience

CSE 571 Artificial Intelligence [Graduate-level Course]

Arizona State University

Tempe, AZ

Fall 2019

- Delivered tutorials on Constraint Satisfaction Problems, PDDL, First Order Logic, and Dynamic Bayesian Networks.
- Co-created the software testbed used to support the course projects. Also created and graded homework assignments and exams.
- Mentored project groups working on class projects.

CS 566 Speech Processing [Graduate-level Course]

Indian Institute of Technology Guwahati

Guwahati, India

Fall 2014

- Co-developed a custom C++ API to use a voice recording tool that students used in course projects.
- Mentored students working on class projects and homeworks.
- Graded homework assignments and exams.

Undergraduate Lab. Courses

Indian Institute of Technology Guwahati

Guwahati, India

2013-2015

- TA for the lab courses: CS 513 Programming Lab.; CS 462 Graphics Lab.; and CS 243 Software Engineering Lab.
- Graded homework assignments and exams.

Academic Projects

Analysis of Gated Updates in Value Iteration Networks

CSE 574 | Planning and Learning Methods in AI | ASU

- Replaced non gated RNN update in Value Iteration Networks with gated updates; long short term memory (LSTM), gated recurrent units (GRUs), and peephole-LSTM.
- Demonstrated that peephole-LSTM performs best for time-bound scenarios, whereas GRUs are recommended for non-time-bound scenarios.

Automatic Image Captioning using Topic Modeling

CSE 569 | Fundamentals of Statistical Learning | ASU

- Generated image caption topics using correspondence Latent Dirichlet Allocation with SIFT image features.
- Used Gibbs sampling to calculate posteriors instead of variational inference.
- Tested on Corel-5K, ESP game, and MIR flickr datasets.
- Showed that the accuracy increases as we increase number of topics till a value, and then starts increasing again.

Artificial Immune System Based Obstacle Avoidance Robot

CS 666 | Mobile Robotics | IIT Guwahati

- Designed a robot which learns to avoid the “dynamic obstacles” using “Clonal Selection” (Immune System concept similar to Reinforcement Learning).
- Experimented with 3 scenarios; 2 robots following fixed paths, 1 robot with one periodically moving obstacle, and 2 robots with unconstrained motion within a region.
- Showed that reward shaping is necessary for robots to learn to avoid obstacles in constrained environments.

Robot Cooperation through Stigmergic Communication

CS 666 | Mobile Robotics | IIT Guwahati

- Modeled the robots to perform sequential tasks by automatic-synchronization using stigmergic communication.
- Used the “Typhon” mobile agent framework to create a two level decentralized “ring” network.
- Sequentially increased the homogeneous cooperating robots in the network till task was completed autonomously.
- Demonstrated the limitations of scalability of “Typhon” which were then addressed in its next version “Tartarus”.

Resource Usage Analysis for Speech Recognition Techniques

Masters Thesis | IIT Guwahati

- Analyzed the impact of i-vectors (a projection of speech feature into total variability space) on devices having low resources like memory, storage space and power for real time speech recognition.
- Showed that i-vectors are better as compared to GMM-UBM and Joint Factor Analysis based approaches for low resource devices.

Predicting Meal Times for Automated Insulin Delivery

CS 565 | Mobile Computing | ASU

- Correlated the noisy real data sampled at different frequencies from continuous glucose monitors (CGMs) and insulin bolus.
- Experimented with SARIMA and RNN for training the time series models for glucose level prediction. Then used the predicted glucose levels and insulin bolus data to infer meal intake times.
- Showed that SARIMA predicts glucose levels with better accuracy than RNN in lesser time.

Smart Analyzer: Predict and Cache Webpages based on Usage

B.E. Capstone Project | SVITS, Indore

- Designed a system that analyzed the patterns of access of websites through a Squid proxy server, and predicted the webpages to be cached.
- Used Apriori algorithm to find correlation between the webpages accessed over different time periods and predict the webpages to be cached depending on the current webpage requested.

Honors & Awards

Jan 2021	GPSA Travel Grant , Arizona State University
Oct 2018	Fulton Schools Experiential Learning Grant , Arizona State University
Aug 2013 - May 2015	Post Graduation Fellowship , All India Council for Technical Education, Government of India

Service

Jan 24 - Jan 25, 2019	Volunteer , Southwest Robotics Symposium 2019, Tempe, AZ ↗
Oct 27 - Nov 02, 2018	Volunteer , 16th Int. Conf. on Principles of Know. Representation & Reasoning, Tempe, AZ ↗
Sep 12 - Sep 14, 2014	Volunteer , National Workshop on GPU Programming and Applications, IIT Guwahati ↗

Other Affiliations

Nov 2020 - (present)	Affiliated Graduate Student , Center for Human-Compatible Artificial Intelligence, UCB ↗
Aug 2018 - (present)	Graduate Student , Autonomous Agents and Intelligent Robots (AAIR) Lab, ASU ↗
Jul 2016 - Dec 2016	Online Contributor , Stanford Scholar Initiative ↗
Jan 2014 - May 2015	Graduate Student , Robotics Lab, IIT Guwahati ↗

References

Prof. Siddharth Srivastava siddharths@asu.edu	Assistant Professor, School of Computing, Informatics, and Decision Systems Engineering, Arizona State University, USA
Prof. Pradip K. Das pkdas@iitg.ac.in	Professor, Department of Computer Science and Engineering, Indian Institute of Technology Guwahati, India
Prof. Shivashankar B. Nair sbnair@iitg.ac.in	Professor, Department of Computer Science and Engineering, Indian Institute of Technology Guwahati, India
